

Battery Container Report

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1 Battery Container Simulation in Ansys

This section presents the finite element analysis (FEA) of a battery container performed using Ansys Mechanical. The objective is to evaluate the structural response of the container under realistic loading conditions, including gravity and acceleration.

1.1 Assumptions

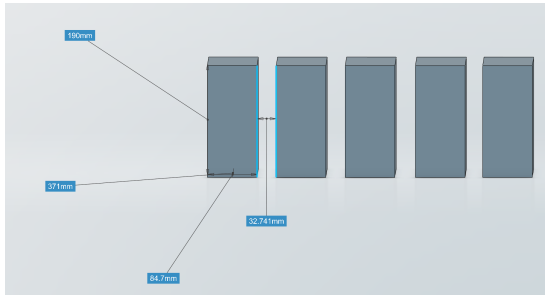
- The battery modules are modeled as simplified rectangular volumes.
- The cells inside each module are not modeled individually.
- Each module is assigned an equivalent material with a density computed as :

$$\rho = \frac{m_{\text{cells}}}{V_{\text{module}}}$$

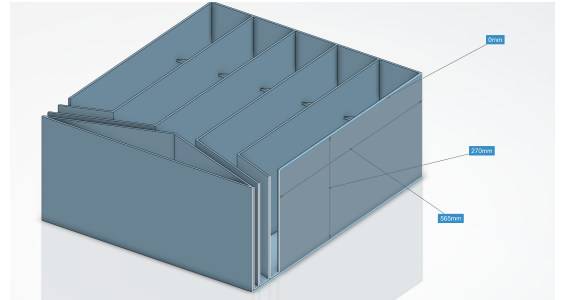
- The weight of the cells is represented using distributed mass and gravity.
- Contact between modules and the container is simplified (bonded assumption).

1.2 Geometry

- The geometry is designed in Onshape and imported into Ansys.
- Container dimensions : $590 \times 670 \times 270$ mm
- Module dimensions : $190 \times 371 \times 84.7$ mm
- Spacing between modules : 32.741 mm
- All components are modeled as rectangular prisms.



(a) Première image



(b) Deuxième image

FIGURE 1 – Deux images côte à côte

1.3 Material Characteristics

The container is made of aluminum to ensure realistic stiffness and strength behavior.

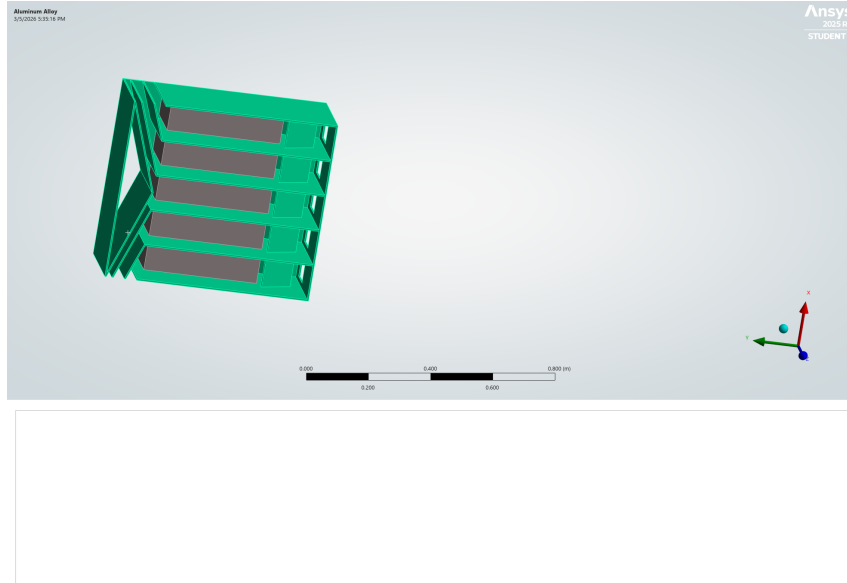


FIGURE 2 – Material properties of aluminum

The modules are assigned an equivalent material based on their density. A material close to this density is selected. In this case, Balsa wood is used with :

$$\rho = 254 \text{ kg/m}^3$$

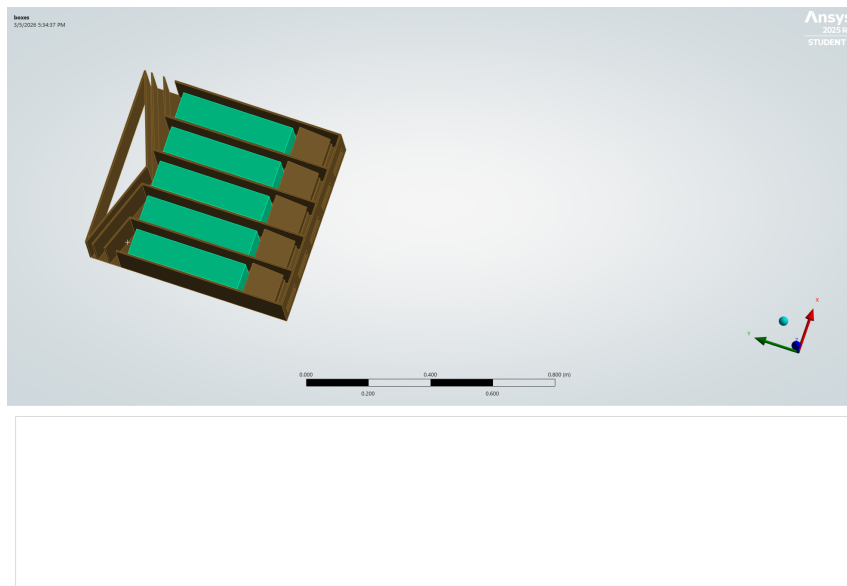


FIGURE 3 – Material properties of aluminum

1.4 Connexion

1.4.1 Contact

The contact is modeled between each module and its corresponding surface on the battery container. The type of contact is **Bonded** because :

- The module is directly screwed to the container.
- There is no gap between the module and the bottom surface.
- There is no sliding between the surfaces.
- The load is transferred through contact pressure.

FIGURE 4 – Connection between module and container

1.5 Loading and Boundary Conditions

1.5.1 Gravity and Mass Representation

The mass of the cells is represented using a distributed mass applied to the module regions.

Gravity is applied as follows :

- Direction : $-Z$
- Magnitude : 9.81 m/s^2

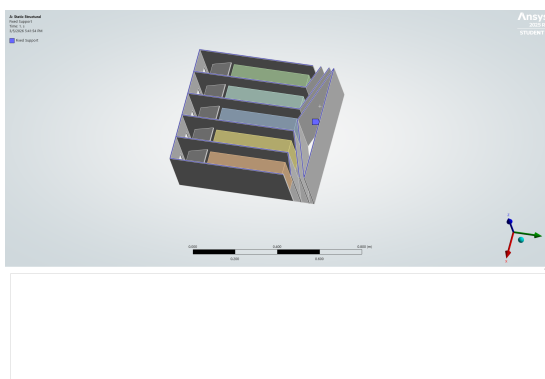
This approach ensures that the weight of the cells is naturally transferred to the bottom surface of the container.

FIGURE 5 – Load transfer to the bottom surface

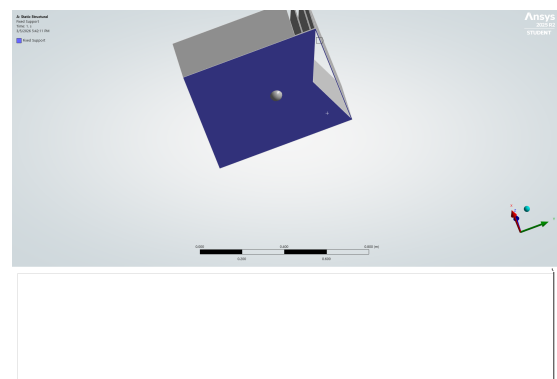
1.5.2 Fixed Supports

Fixed supports are applied to represent the attachment of the container to the vehicle structure :

- Along the upper mounting edges
- On selected bottom regions



(a) Première image



(b) Deuxième image

FIGURE 6 – Boundary conditions (fixed supports)

1.5.3 Acceleration Loads

Acceleration loads are applied to simulate extreme operating conditions :

- $40g$ in the longitudinal direction (x)
- $40g$ in the lateral direction (y)
- $20g$ in the vertical direction (z)

1.6 Meshing Strategy

First simulation is done with the default meshing A refined mesh is applied in critical areas, especially on the bottom surface where stress concentrations are expected.

- Element size on bottom surface : 0.01 m



FIGURE 7 – Mesh configuration

1.7 Results

1.7.1 Total Deformation

The total deformation highlights the global stiffness of the structure, with maximum deformation occurring in unsupported regions. The maximum deformation is $9.23 \times 10^{-6}\text{ m}$ ($9.23\text{ }\mu\text{m}$), which is acceptable for this application. For acceleration along the x -direction, deformation occurs primarily on the lateral side. In reality, this lateral side will be maintained fixed by connection to another component, so this aspect requires further detailed study.

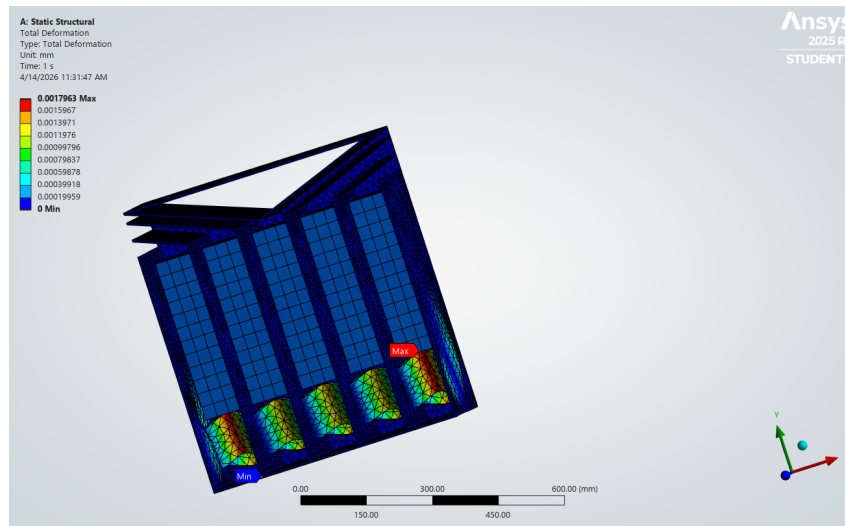


FIGURE 8 – Total deformation

1.7.2 Directional Deformation

This is a useful plot because it shows the deformation along the X axis, not just the total displacement. The values are very small, from about -0.000503 mm to +0.000503 mm, so the part is only moving slightly in the global X direction

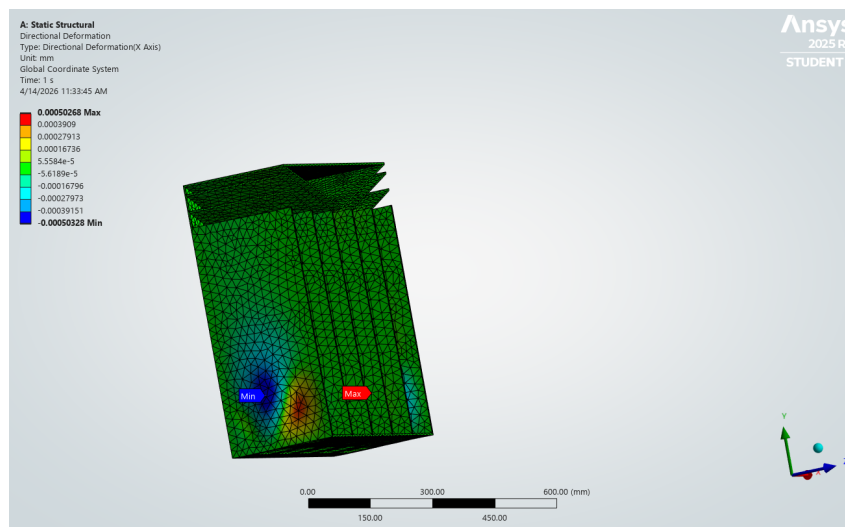


FIGURE 9 – Directional deformation

1.7.3 Equivalent (Von Mises) Stress

This equivalent von Mises stress result looks very low overall : the legend shows a maximum of about 0.36969 MPa and a minimum near zero, so the part is far from yielding. Because the peak stress is only around 0.37 MPa and the fact that the material is aluminum the model is in a very safe elastic range

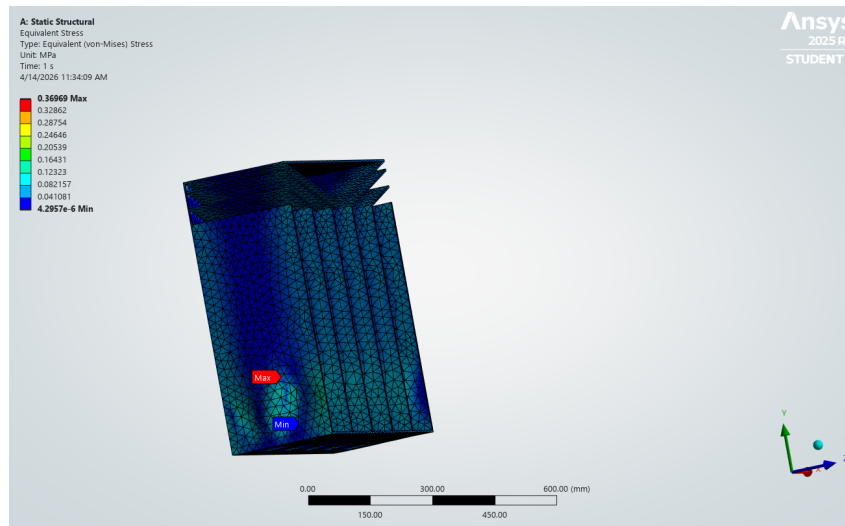


FIGURE 10 – Equivalent (Von Mises) stress

2 Conclusion

The overall results are good since the deformations are very low. The results also fit well with the characteristics of aluminum. For a more accurate simulation, we will modify the surface of the module in contact with the bottom surface of the container to improve the transmission of forces between the two parts.

Thank you for your attention.